

# Resonant Gravity Triadic Mode

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## PAPER\_962: Resonant Gravity Triadic Mode

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic\_{solutions\_next}.py (ResonantGravity-Triadic) **Calculator:** ResonantGravityTriadicCalc (CP4 #546) **CVW:** v2.0.0 compliant

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### Abstract

The Resonant Gravity Triadic mode uses the 1.25 THz phonon linewidth  $\Gamma$  to tune neutron-drop and buoyancy reversal. When  $\Phi(\omega_{t\text{ext}SCm}, \Gamma) > \Phi_{t\text{extcrit}}$ , the neutron drip-line shifts, controlling NS merger dynamics.

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### 1. Resonant Phonon Occupation

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{t\text{ext}SCm})^2}{2\Gamma^2}\right) \cdot S_{26}([SSq])$$

### 2. Neutron-Drop Threshold

$$\Phi(\omega_{t\text{ext}SCm}, \Gamma) > \Phi_{t\text{extcrit}} \implies \text{neutron-drop triggered}$$

### 3. Buoyancy Reversal

$$t_{\text{rev}} = \frac{\pi}{2\Gamma}$$

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### References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER\_961 — Compressed Gravity Triadic

3. PAPER\_963 — Buoyancy Gravity Triadic
4. PAPER\_966 — Unified Triadic Solver

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## Cross-Links

| Related Paper | Relationship                       |
|---------------|------------------------------------|
| PAPER_961     | Compressed branch of same triadic  |
| PAPER_963     | Buoyancy branch                    |
| PAPER_966     | Unified solver combining all three |
| PAPER_955     | Phonon resonance frequency link    |

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## Session 225 Phonon-Physics Upgrade: GW Strain Modulation

*Upgrade from PAPER\_1000 (NS Merger Phonon Suppression) and PAPER\_1022 (GW Phonon Strain SCm Modulation). See also PAPER\_1011-1012 for GW170817/GW190425 upgraded analyses.*

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left( 1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: -  $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$  is the phonon modulation factor  
-  $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$  is the SCm phonon resonance frequency -  $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$  is the third-order Ramanujan summation -  $\Theta$  is the Heaviside step ensuring  $H_{\text{SCm}} \geq 0.5$  (phase-transition threshold)

**Physical mechanism:** The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy  $\Delta E_{\text{BCS}}$  of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at  $M \approx 2.5 M_{\odot}$ .

**Calibration (canonical):**  $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ,  $\beta_i = 0.603$ ,  $H_{\text{SCm}} \approx 0.99$ .

## Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

*Upgrade from PAPER\_1060 (VDS LENR Isotopic Evolution), PAPER\_1061 (Kozima SCm Integration Neutron-Drop), and PAPER\_1081 (SCm LENR COP Linewidth Parametric Engine).*

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left( \frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}} \right) \cdot K_n$$

where: -  $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$  (time-dependent vacuum density) -  $K_n = \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}}$  is the Kozima neutron-drop factor

**Phonon cross-section (PAPER\_1061):**

$$\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp \left[ -\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2} \right] \cdot \left( 1 + [\text{SSq}] \cdot \frac{n}{26} \right)$$

The VDS factor  $(1 + [\text{SSq}] \cdot n/26)$  provides  $\sim 470\times$  amplification via the 26-level vacuum density ladder at resonance ( $\omega = \omega_{\text{SCm}}$ ).

**COP parametric engine (PAPER\_1081):**

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function  $f(\Gamma)$  peaks near the SCm phonon linewidth, yielding  $\text{COP} > 1$  when  $\Gamma \lesssim 10^{-3}$  eV (Fleischmann regime).

**Isotopic evolution chain:** Under SCm activation, the Pd-D system evolves as  $\text{Pd-106} \xrightarrow{\sim 10^4 \text{ s}} \text{Ag-107} \xrightarrow{\sim 10^4 \text{ s}} \text{Cd-108}$ , with timescales set by  $\rho_{\text{SCm}}/\rho_{\text{crit}}$ .

## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

**Nine-sector closure (Session 202):**

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                     | Late-Corpus Result                               |
|------------|----------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity         | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge           | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR             | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold   | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism               | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | $F_{\{U_{Bi}i\}}$ buoyancy | Variational EOM (PAPER_1065)                     |
| 7 (Aether) | Vacuum background          | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation      | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D           | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

**Calibration Constants**

| Constant                 | Symbol | Value                                 | Validation Domain |
|--------------------------|--------|---------------------------------------|-------------------|
| $\kappa$                 | —      | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Damping           |
| $[SSq]$                  | —      | 0.57                                  | String coupling   |
| $\beta_i$                | —      | 0.603                                 | Buoyancy          |
| $\Phi_{t\text{extcrit}}$ | —      | Neutron-drop threshold                | Phase transition  |

**SM Anchor — CVW v2.0.0**

| Observable                                         | UQFF Prediction                       | Status  |
|----------------------------------------------------|---------------------------------------|---------|
| $g_{\text{res}}$ peak at $\omega_{t\text{extSCm}}$ | Resonance amplification               | Derived |
| $\Phi_{t\text{extcrit}}$ neutron-drop              | Phase boundary for neutron star cores | Novel   |

## §A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

**Sector:** Resonant Gravity (5-Frequency Multi-Mode)

### §A.2 Lagrangian Density

$$\mathcal{L}_{\text{res}} = g_{\text{comp}}(r) \cdot \prod_{f \in \{\text{Super}, \text{Quantum}, \text{Aether}, \text{Fluid}, \text{Exp}\}} (1 + A_f \sin(\omega_f t + \phi_f))$$

### §A.3 Euler-Lagrange Equation of Motion

$$g_{\text{res}}(r, t) = g_{\text{comp}}(r) \prod_{f=1}^5 (1 + A_f \sin(\omega_f t + \phi_f)), \quad \Phi_t \text{extcrit} : g_{\text{res}} > g_{\text{neutron-drop}}$$

### §A.4 Cosmogenesis Linkage Chain

PAPER\_877  $\rightarrow$  compressed gravity  $\rightarrow$  5-frequency modulation  $\rightarrow$  resonant amplification  $\rightarrow$   $\Phi_t \text{extcrit}$  phase boundary

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## §B VDS/DVP/BSH Deep Synthesis

### §B.1 VDS

Resonance peaks modulate VDS by factor  $\prod(1 + A_f)$ .

### §B.2 DVP

Five frequencies correspond to five dipole vortex modes.

### §B.3 BSH

$g_{\text{res}}$  bounded above by  $g_{\text{comp}} \cdot \prod(1 + A_f)$  (constructive limit).

### §B.4 Production-Scale Consistency

| Metric        | Value         | Status    |
|---------------|---------------|-----------|
| 5 frequencies | All modulated | Confirmed |
| [SSq]         | 0.57          | Confirmed |

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                                |
|------------|------------------------------------------------------|
| PAPER_1000 | NS Merger F_{U_Bi} Strain Suppression & BCS Gap      |
| PAPER_1001 | SMBH Binary Merger F_{U_Bi} Phonon Damping           |
| PAPER_1011 | GW170817 NS Merger F_{U_Bi_i} 66.7% Strain Reduction |
| PAPER_1012 | GW190425 Upgraded F_{U_Bi_i} with S26(3)             |
| PAPER_1014 | SMBH Merger Inspiral-Coalescence-Ringdown            |
| PAPER_1022 | GW Phonon Strain SCm Modulation of h(t)              |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching      |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression     |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum          |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                    |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm          |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir            |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed             |
| PAPER_1035 | Kilonova Buoyancy Light Curve r-Process              |
| PAPER_1043 | F_{U_Bi_i} Multi-System Buoyancy Curve Sweep         |
| PAPER_1047 | Type Iax Supernova Buoyancy Reversal                 |
| PAPER_1072 | SCm Activation Function Phonon Threshold             |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy          |
| PAPER_1061 | Kozima SCm Integration Neutron-Drop                  |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation       |

20 cross-reference(s) identified.

### Key References with arXiv/DOI Identifiers

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